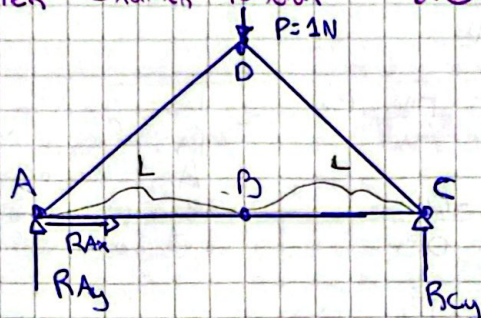


Primer Examen Pascal

10/5/2024

1)



a) Determinar el Valor de las Reacciones

$$\sum F_x = 0$$

$$[R_{Ax} = 0]$$

$$\sum F_y = 0$$

$$R_{Ay} + R_{Cy} - P = 0$$

$$R_{Ay} = P - R_{Cy}$$

$$R_{Ay} = 1 - 0.5$$

$$[R_{Ay} = 0.5 \text{ N}]$$

Tomamos el punto A para momento

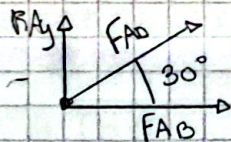
$$\sum M = 0$$

$$-R_{Cy} \cdot 2L + P \cdot L = 0$$

$$-R_{Cy} = \frac{-P \cdot L}{2L}$$

$$[R_{Cy} = \frac{1}{2} = 0.5 \text{ N}]$$

b) Determinar el esfuerzo sobre la barra A-B



$$\sum F_x = 0$$

$$F_{AB} + F_{AD} \cdot \cos 30^\circ = 0$$

$$F_{AB} = -F_{AD} \cos 30^\circ$$

$$F_{AB} = -1 \text{ N} \cdot \cos 30^\circ$$

$$[F_{AB} = -\frac{\sqrt{3}}{2}]$$

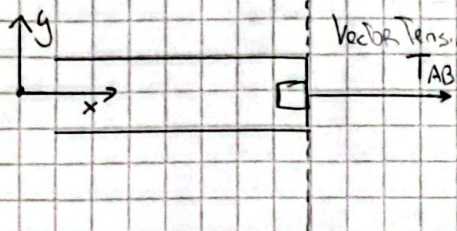
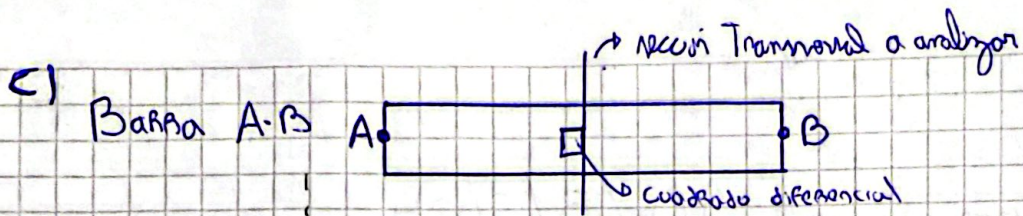
esfuerzo sobre barra A-B

$$\sum F_y = 0$$

$$R_{Ay} + F_{AD} \cdot \sin 30^\circ = 0$$

$$F_{AD} = \frac{-R_{Ay}}{\sin 30^\circ} = \frac{-0.5 \text{ N}}{\sin 30^\circ}$$

$$F_{AD} = -1 \text{ N}$$



$$T_{AB} = \frac{F_{AB} \cdot e_x}{A_{AB}}$$

$$T_{AB} = \frac{-\sqrt{3}/2}{0.5}$$

$$T_{AB} = -\sqrt{3}$$

$$T_{AB} = \text{Tensión en AB}$$

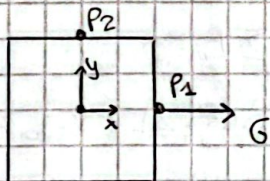
$$F_{AB} = \text{Fuerza en AB}$$

$$A_{AB} = \text{Área de AB}$$

$$e_x = \text{normal a la dirección}$$

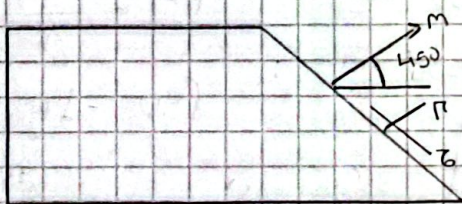
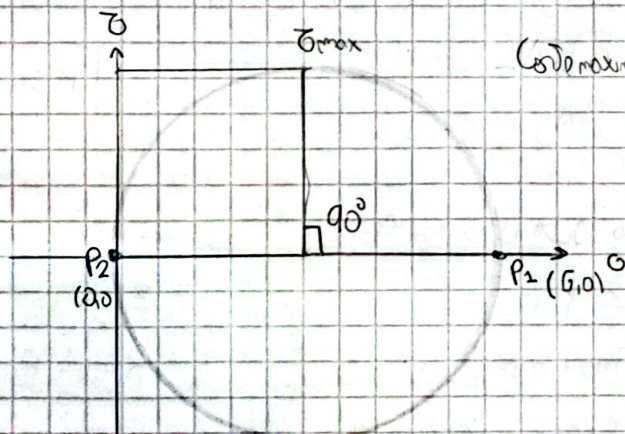
$$\text{modo positivo elástico en el eje x}$$

Cuadrado Diferencial



$$P_1 = (0,0)$$

$$P_2 = (0,1)$$



2) a) Los vectores a, b y c son campos vectoriales (as de un tensor de orden 1 dependientes de las coordenadas del dominio) y que se cumple que $a \cdot c = 0$. demuestramos

$$\nabla \cdot [a \times (b \times c)] = -b[(\nabla a) \cdot c + (\nabla \cdot c)a] - a(\nabla b) \cdot c$$

avanzamos por la derecha

$$\frac{\partial}{\partial x_i} [a \times (b \times c)]_i = \frac{\partial}{\partial x_i} [\epsilon_{ijk} a_j (b \times c)_k] = \frac{\partial}{\partial x_i} [\epsilon_{ijk} a_j \epsilon_{kmn} b_m c_n]$$

$$\frac{\partial}{\partial x_i} [\epsilon_{ijk} \epsilon_{kmn} a_j b_m c_n] \quad \text{notamos indices } ijk$$

$$\frac{\partial}{\partial x_i} [\epsilon_{kij} \epsilon_{kmn} a_j b_m c_n] \quad \text{cambiamos propiedad}$$

$$\frac{\partial}{\partial x_i} [S_{im} S_{jn} - S_{jm} S_{in}] a_j b_m c_n \quad \text{multiplicamos}$$

$$\frac{\partial}{\partial x_i} [S_{im} S_{jn} a_j b_m c_n - S_{jm} S_{in} a_j b_m c_n]$$

$$\frac{\partial}{\partial x_i} [b_i (\underbrace{a_m c_m}_0) - a_m c_i b_m] \quad [a_i c_i = 0]$$

$$\frac{\partial}{\partial x_i} [b_i \cdot 0 - a_m c_i b_m]$$

$$\frac{\partial}{\partial x_i} [-a_m c_i b_m] = -\frac{\partial a_m c_i b_m}{\partial x_i} - \frac{\partial c_i a_m b_m}{\partial x_i} - \frac{\partial b_m a_m c_i}{\partial x_i}$$

$$= -(\nabla a) \cdot c \, b - (\nabla c) \cdot a \, b - (\nabla b) \cdot a \, c$$

$$= -b[(\nabla a) \cdot c + (\nabla c) \cdot a] - a(\nabla b) \cdot c$$

✓

b) Considerando que σ es un tensor de Tercer orden demuestre

$$\epsilon_{ijk} S_{mk} \sigma_{jm} = 0$$

$$\frac{1}{2} \epsilon_{ijk} S_{mk} \sigma_{jm} + \frac{1}{2} \epsilon_{ijk} S_{mk} \sigma_{jm} \quad \text{doble cantidad}$$

$$\frac{1}{2} \epsilon_{ijk} S_{mk} \sigma_{jm} - \frac{1}{2} \epsilon_{ikj} S_{mk} \sigma_{jm} \quad \text{permutar indices}$$

$$\frac{1}{2} [\epsilon_{ijk} S_{mk} \sigma_{jm} - \epsilon_{ikj} S_{mk} \sigma_{jm}] \quad \text{contrange}$$

$$\frac{1}{2} [\epsilon_{ijk} \sigma_{jk} - \epsilon_{ikj} \sigma_{jk}] \quad \text{Como es Tensor de Tercer orden}$$

$$\sigma_{jk} = \sigma_{kj}$$

$$\frac{1}{2} [\underbrace{\epsilon_{ijk} \sigma_{jk}}_{\text{iguales}} - \underbrace{\epsilon_{ikj} \sigma_{kj}}_{\text{iguales}}]$$

$$\frac{1}{2} \cdot 0 = 0 \quad \checkmark$$

$$3) \quad \theta = \begin{bmatrix} X_1^2 X_2 & (1-X_2^2)X_1 & 0 \\ (1-X_2^2)X_1 & \frac{X_1^3 - 3X_2}{3} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

a) distribuição de Laplace de mediana zero que se comporta como uma de equação

$$\frac{\partial \theta_{ij}}{\partial X_i} = -b_i$$

$$\begin{cases} \frac{\partial \theta_{11}}{\partial X_1} + \frac{\partial \theta_{12}}{\partial X_2} + \frac{\partial \theta_{13}}{\partial X_3} + b_1 = 0 & \textcircled{1} \quad b_1, b_2, b_3? \\ \frac{\partial \theta_{21}}{\partial X_1} + \frac{\partial \theta_{22}}{\partial X_2} + \frac{\partial \theta_{23}}{\partial X_3} + b_2 = 0 & \textcircled{2} \\ \frac{\partial \theta_{31}}{\partial X_1} + \frac{\partial \theta_{32}}{\partial X_2} + \frac{\partial \theta_{33}}{\partial X_3} + b_3 = 0 & \textcircled{3} \end{cases} \quad \text{para } p_1 \text{ de mediana}$$

$$\textcircled{1} \quad 2X_1 X_2 + (-2X_2 X_1) + 0 + b_1 \rightarrow [b_1 = 0]$$

$$\textcircled{2} \quad (1-X_2^2) + \left(\frac{3X_2^2 - \frac{1}{3}}{3}\right) + 0 + b_2 \rightarrow [b_2 = 0]$$

$$\textcircled{3} \quad 0 + 0 + 0 + b_3 \rightarrow [b_3 = 0]$$

b) Tensões Principais em el emo $P(a, 0, 2\sqrt{a})$ $X_1 = a$ $X_2 = 0$ $X_3 = 2\sqrt{a}$

$$\theta = \begin{bmatrix} a^2 \cdot 0 & (1-0^2)a & 0 \\ (1-0^2)a & (0^3 - 3 \cdot 0^2)/3 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & a & 0 \\ a & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\det \theta = \begin{vmatrix} 0-\lambda & a \\ a & 0-\lambda \end{vmatrix}$$

Para los Tensores principales buscamos los autovalores

$$(-\lambda - \lambda) - (a \cdot a)$$

$$+\lambda^2 - a^2 = 0 \quad \left\{ \begin{array}{l} a=1 \\ b=0 \\ c=-a^2 \end{array} \right. \left[\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \right]$$

$$-0 \pm \sqrt{0^2 - 4 \cdot 1 \cdot (-1)}$$

$$\pm \sqrt{\frac{+4a^2}{2}}$$

$$\pm \frac{2a}{2} \rightarrow \begin{cases} +a & \sigma_{\max} \\ -a & \sigma_{\min} \end{cases}$$

c) máxima Tensión de corte

$$\tau_{\max} = \frac{\sigma_{\max} - \sigma_{\min}}{2} = \frac{a - (-a)}{2} = \frac{2a}{2} = a$$

d) Vector Tensión que corta sobre el plano $X_1 + 3X_2 + X_3 = \sqrt{a}(\sqrt{a}+2)$

Tornteo normal que apunta al primer cuadrante. [Puntos cuadrante primeros positivos]

Sacamos el VF para obtener la normal $aX_1 + bX_2 + cX_3 = 1X_1 + 3X_2 + 1X_3$

$$\nabla F = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} \quad \text{cons. no quedan variables, no evaluamos en el punto}$$

$$m = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} \quad \hat{m} = \frac{1}{\sqrt{1^2 + 3^2 + 1^2}} \cdot \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{11}} \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix}$$

$$T = \sigma \cdot \hat{m} = \begin{bmatrix} 0 & a & 0 \\ a & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \cdot \frac{1}{\sqrt{11}} \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{11}} \begin{pmatrix} 3a \\ a \\ 0 \end{pmatrix}$$

Regla del Cociente: Si el Producto interno entre una entidad matemática de Componentes desordenados y un Tensor arbitrario de rango concreto da siempre como resultado un Tensor de rango concreto entonces la Entidad es obligatoriamente un Tensor.

$T_{ij} = \Theta_{ij} M_j$ aplico regla del cociente para verificar Θ_{ij} sabiendo que T_{ij} y M_j son Tensores de rango 1

$$T'_{ij} = \Theta'_{ij} M'_j$$

$$Q_{ip} T_p = \Theta'_{ij} M'_j$$

$$Q_{ip} \Theta_{ps} M_s = \Theta'_{ij} Q_{js} M'_s$$

$$Q_{ip} \Theta_{ps} = \Theta'_{ij} Q_{js}$$

$$Q_{ip} \Theta_{ps} Q_{ks} = \Theta'_{ij} Q_{js} Q_{ks}$$

$$Q_{ip} \Theta_{ps} Q_{ks} = \Theta'_{ij} S_{jk}$$

$$\Theta_{ps} Q_{ip} Q_{ks} = \Theta'_{jk}$$

Tensor de rango 2

$$T'_i = Q_{ip} T_p$$

$$M'_j = Q_{js} M_s$$

$$T_p = \Theta_{ps} M_s$$

M_s es arbitrario se cumple siempre

multiplicamos por Q_{ks}

Q_{ks} denotamos por T_{kij} es ortogonal

Matriz ortogonal x Su Transpuesta = Matriz de Kronecker

$$Q_{js} Q_{ks} = S_{jk}$$